Loss Functions

15 December 2025 19:39

Loss function

ML → Loss ← OLS, GD

$\frac{dL}{dw} = 0$

$\boxed{\frac{dL}{dw}}$

$w = w - \eta \frac{dL}{dw}$

(I) Regression
- → MSE
- → MAE
- → Number Loss

(II) Classification
- → Binary Cross Entropy
- → Categorical " "
- → Hingeloss " "

Loss Func	Cost Func
↓ single row error	Data error.

Regression Error Functions MSE

15 December 2025 19:45

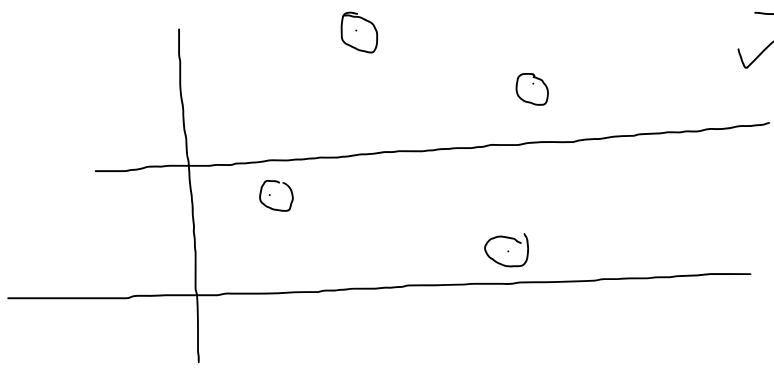
MSE (Mean Sq. Error) / L_2 Loss / Sq. Loss :-

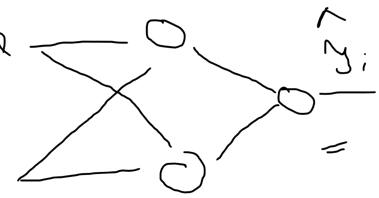
$$L = (y_i - \hat{y}_i)^2$$

$$C = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

(L)

x_1 cgpa	x_2 iq	y Package	$\hat{y}_i$ Pred
6.3	100	6.3	6.1
7.1	91	4.1	4
8.5	83	3.5	3.7
9.2	102	7.2	7
✓ 4	70	12	3

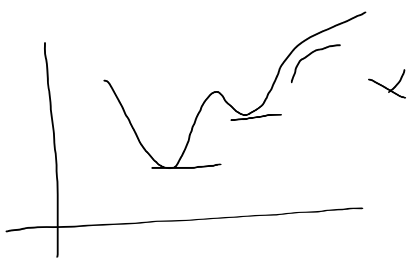




Adv

→ Diffⁿ

⇒ One local minima

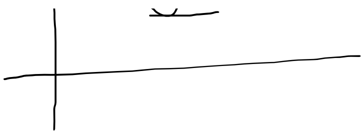


Dis Adv

i) Sq. Unit

ii) Not Robust to Outliers

$$MSE = \frac{1}{n}\sum(Y_{true} - Y_{pred})^2$$



$$\begin{bmatrix} 0.2 & 0.1 & 0.2 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 0.04 & 0.01 & 0.04 & 81 \end{bmatrix}$$

Unstable Gradient:

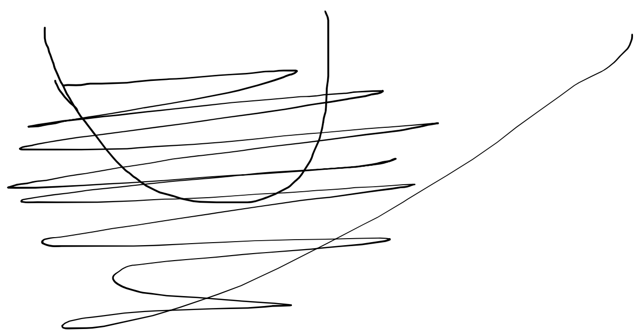
$$\frac{dL}{dy_{pred}} \Big/ \frac{dL}{d\hat{y}}$$

GD

$$w = w - \eta \frac{dL}{dw}$$

$$L = f(\hat{y}_1)$$

y_{pred}



$$L = (y_i - \hat{y}_i)^2 \quad \Big| \quad (\hat{y}_i - y_i)^2$$

$$\frac{dL}{d\hat{y}} = \frac{2(\hat{y}_i - y_i)}{}$$

$$\boxed{\begin{matrix} 0.1 & 0.01 \\ 2 & 4 \end{matrix}}$$

Regression Error Functions MAE

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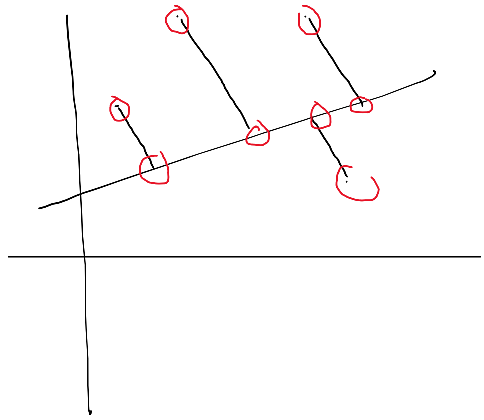
$$L = |y_i - \hat{y}_i|$$



$$C = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

Adv

i) Same Unit

ii) Outlier robust

Dis adv

→ Not Diff^n at $x=0$

$$\lim_{h \to 0^+} \frac{f(x+h)-f(x)}{h} = \lim_{h \to 0^-} \frac{f(x)-f(x-h)}{h}$$

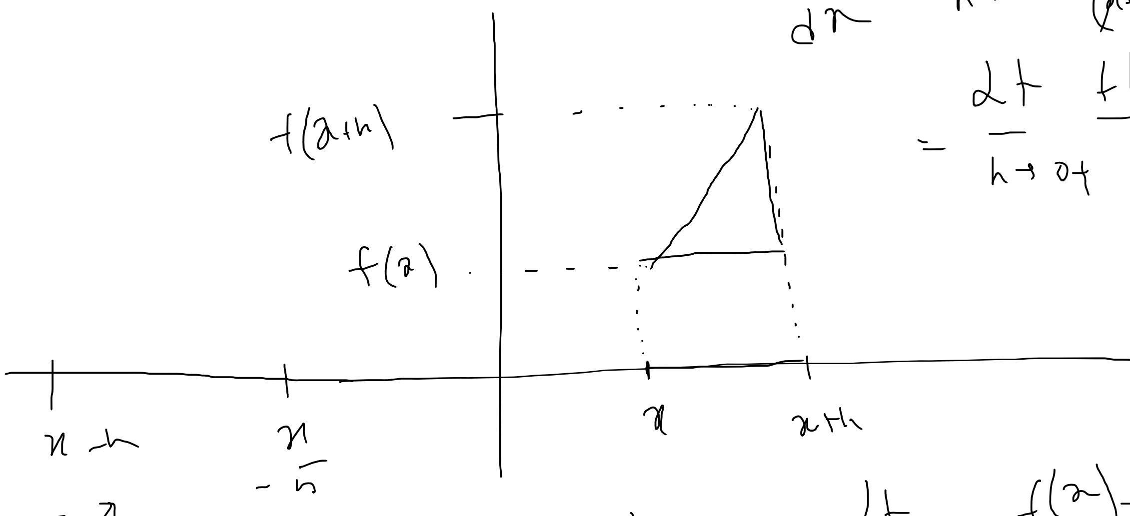
$$\Downarrow \qquad\qquad \Downarrow$$

$$+1 \neq -1$$

$|x|$ not diff^n at $x=0$

$$\frac{dy}{dx} = \lim_{h \to 0^+} \frac{f(x+h)-f(x)}{(x+h)-x}$$

$$= \lim_{h \to 0^+} \frac{f(x+h)-f(x)}{h}$$



$$\lim_{h \to 0^-} \frac{f(x)-f(x-h)}{x-(x-h)} = \lim_{h \to 0^-} \frac{f(x)-f(x-h)}{h}$$

$$\lim_{h \to 0^-} \frac{\quad}{x - (x-h)} \qquad h \to 0^-$$

$h - x$

$-5 + 7$
$= 2$

$$|x| = +x$$

$$\lim_{h \to 0^+} \frac{f(x+h) - f(x)}{h}$$

$$\boxed{-5 - 7 = -12}$$

$h \to 0^+$

$$\lim_{h \to 0^-} \frac{|x| - |x-h|}{h}$$

$$= \frac{x - x + h}{h}$$

$$= \lim_{h \to 0^+} \frac{|x+h| - |x|}{h}$$

$$= \lim_{h \to 0^+} \frac{\cancel{x} + h - \cancel{x}}{h}$$

$$= \lim_{h \to 0^+} \frac{h}{h} = +1$$

$|x|$

-1

Huber Loss

15 December 2025 20:12

$$L = \begin{cases} \frac{1}{2}(y-\hat{y})^2 \leftarrow \text{MSE} & \text{for } |y-\hat{y}| \le \delta \\ \delta|y-\hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases}$$

↑ MAE

↑ hyper param

Binary Cross Entropy

15 December 2025 20:16

Log Loss

→ classification

→ Two classes

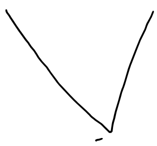
$$L = -y_i \log(\hat{y}_i) - (1-y_i)\log(1-\hat{y}_i)$$

$$C = -\frac{1}{n}\left[\sum_{i=1}^{n}\left[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right]\right]$$

Adv

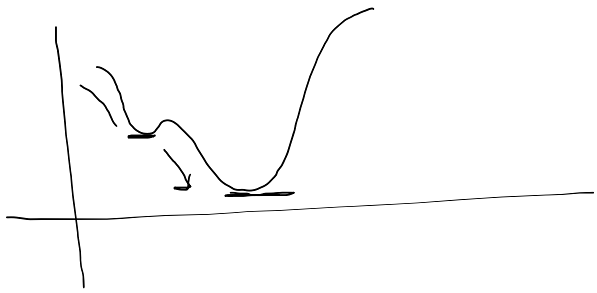
1. Diff^n

$$\boxed{\frac{dL}{dW}}$$



Disadv

i) Multiple local minima



→ Multiclass Classification

$$L = -\sum_{j=1}^{k} y_j \log(\hat{y}_j)$$

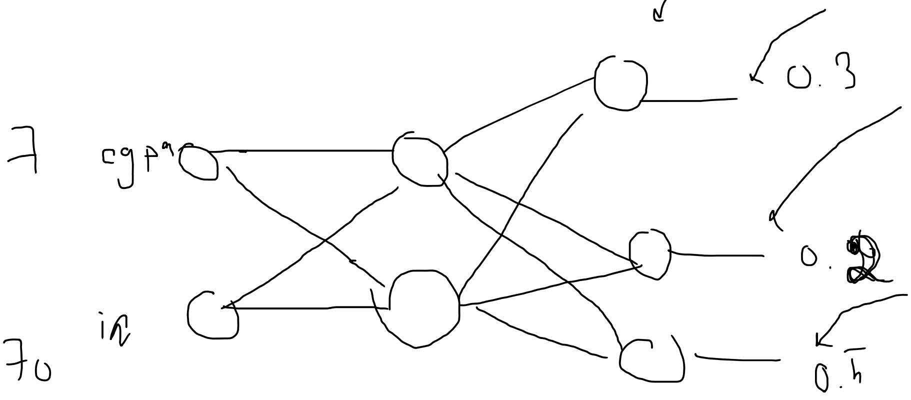
K → # classes

activation = 'softmax'

$$\sigma(z_1) = \frac{e^{z_1}}{e^{z_1}+e^{z_2}+e^{z_3}}$$

$$\sigma(z_2) = \frac{e^{z_2}}{e^{z_1}+e^{z_2}+e^{z_3}}$$

$$\sigma(z_3) = \frac{e^{z_3}}{e^{z_1}+e^{z_2}+e^{z_3}}$$



[0.3 0.2 0.5]

[0 0 1]

$\sigma(z_1)+\sigma(z_2)+\sigma(z_3)=1$

$$= -\underbrace{y_1}_{0}\log\hat{y}_1 - \underbrace{y_2}_{0}\log\hat{y}_2 - \underbrace{y_3}_{1}\log\hat{y}_3$$

$$= -\boxed{\log(0.5)}$$

$= +0.3$

cgpa	iq	placement
8	80	yes
6	60	no
7	70	maybe

OHE →

Yes	No	Maybe
1	0	0
0	1	0
0	0	1

Sparse Categorical Cross Entropy

15 December 2025 20:28

Catg. Cross Ent ← OHE

Sparse CCE ← Integer Encoding

Conclusion

15 December 2025 20:33

Regression → Outlier
- Present → MAE
- Absent → MSE

} huber loss

Classification
- → binary → BCE
- → Multiclass
  - → CCE
  - → SCCE